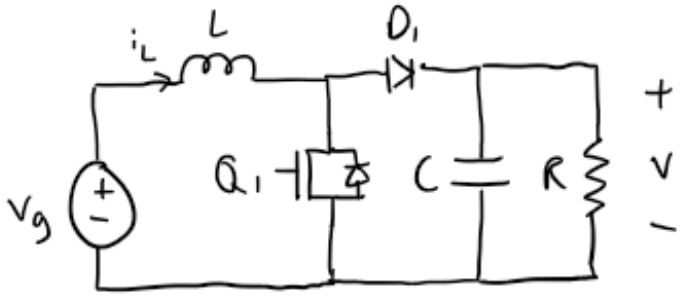
Erickson 7.1/8.21: The Boost Converter



$V_s = 48V,\ L = 100\mu H,\ C = 33\mu F,\ f_{sw} = 200kHz$

$R = 12\Omega$

a) Find the averaged model: CCM: 2-states

State 1: Q_1 ON

$$v_L = \langle v_g \rangle$$

$$i_C = -\frac{\langle v \rangle}{R}$$

$0 < t < d \cdot T_s$

State 2: Q_1 OFF

$$v_L = \langle v_g \rangle - \langle v \rangle$$

$$i_C = i_L - \frac{\langle v \rangle}{R}$$

$d \cdot T_s < t < T_s$

$$\langle v_L \rangle_{T_s} = \frac{1}{T_s} \cdot \int_0^{T_s} v_L \, dt$$

$$\langle v_L \rangle = \frac{1}{T_s} \cdot \left[\int_0^{dT_s} \langle v_g \rangle dt + \int_{dT_s}^{T_s} \langle v_g \rangle - \langle v \rangle \, dt \right]$$

$$\langle v_L \rangle = \frac{1}{T_s} \cdot \left[dT_s \langle v_g \rangle + T_s \langle v_g \rangle - T_s \langle v \rangle - dT_s \langle v_g \rangle + dT_s \langle v \rangle \right]$$

$$\langle v_L \rangle = \langle v_g \rangle - (1-d) \langle v \rangle \quad (1)$$

$$\langle i_L \rangle_{T_s} = \frac{1}{T_s} \int_0^{T_s} i_L \, dt$$

$$\langle i_C \rangle = \frac{1}{T_s} \cdot \left[\int_0^{dT_s} -\frac{\langle v \rangle}{R} dt + \int_{dT_s}^{T_s} \langle i_L \rangle - \frac{\langle v \rangle}{R} \, dt \right]$$

$$\langle i_C \rangle = \frac{1}{T_s} \cdot \left[-\frac{dT_s \langle v \rangle}{R} + T_s \langle i_L \rangle - T_s \frac{\langle v \rangle}{R} - dT_s \langle i_L \rangle + \frac{dT_s \langle v \rangle}{R} \right]$$

$$\langle i_C \rangle = -\frac{\langle v \rangle}{R} + (1-d) \langle i_L \rangle \quad (2)$$

$$\langle v_L \rangle = \langle v_g \rangle - (1-d)\langle v \rangle \quad (1)$$

$$\langle v_L \rangle = L \frac{d\langle i_L \rangle}{dt}$$

$$\boxed{L \frac{d\langle i_L \rangle_{T_s}}{dt} = \langle v_g \rangle - \underbrace{(1-d)}_{d'}\langle v \rangle}$$

$$\langle i_C \rangle = -\frac{\langle v \rangle}{R} + (1-d)\langle i_L \rangle \quad (2)$$

$$\langle i_C \rangle = C \frac{d\langle v \rangle_{T_s}}{dt}$$

$$\boxed{C \frac{d\langle v \rangle_{T_s}}{dt} = -\frac{\langle v \rangle}{R} + \underbrace{(1-d)}_{d'}\langle i_L \rangle}$$

What terms makes the averaged model of the boost converter non-linear?

Answer: $d' \cdot \langle v \rangle$ & $d' \cdot \langle i_L \rangle$

because the products above are each made up of two time-varying signals.

If d changes, so does $\langle v \rangle$, so its non-linear.

b) Perturb & Linearize to create a linear model

$\langle v_g \rangle = V_g + \hat{v}_g$, $d = D + \hat{d}$, $d' = 1 - D - \hat{d}$

$\langle i_L \rangle = I_L + \hat{i}_L$, $\langle v \rangle = V + \hat{v}$

$V_g = 48\,V$, $D = 0.6$, $I_L = \frac{V}{D' \cdot R} = \frac{V_g / D'}{D' \cdot R} = \frac{V_g}{D'^2 R} = 25A$

$V = \frac{V_g}{D'} = \frac{48}{0.4} = 120V$ $\quad D' = 0.4$

$$L \frac{d\langle i_L \rangle}{dt} = \langle v_g \rangle - d' \langle v \rangle \quad (1)$$

$$L \cdot \frac{d}{dt}\left[I_L + \hat{i}_L\right] = V_g + \hat{v}_g - \left(D' - \hat{d}\right)\left(V + \hat{v}\right)$$

$\frac{d}{dt} \cdot I_L = 0$: DC-value

$$L \cdot \frac{d}{dt}\hat{i}_L = V_g + \hat{v}_g - D'V + \hat{d}V - D'\hat{v} + \hat{d}\hat{v}$$

→ 2nd-order ignore based on small-ripple approximation

$$\frac{d}{dt}\hat{i}_L = \frac{V_g - D'V}{L} + \frac{1}{L}\hat{v}_g + \frac{V}{L}\hat{d} - \frac{D'}{L}\hat{v}$$

$$\frac{48 - 0.4 \cdot 120}{L} = 0$$

$$\frac{d}{dt}\hat{i}_L = \frac{1}{L}\hat{v}_g + \frac{V}{L}\hat{d} - \frac{D'}{L}\hat{v} \quad (3)$$

$$C\frac{d}{dt}\langle v\rangle = \frac{-\langle v\rangle}{R} + d'\langle i_L\rangle \quad (2)$$

$$C\frac{d}{dt}\left[V+\hat{v}\right] = \frac{-V}{R} - \frac{\hat{v}}{R} + \left(D'-\hat{d}\right)\left(I_L+\hat{i}_L\right)$$

$$C\frac{d}{dt}\hat{v} = \frac{-V}{R} - \frac{1}{R}\hat{v} + D'I_L + D'\hat{i}_L - I_L\hat{d} - \hat{d}\hat{i}_L$$

→ 2nd-order. Ignore ∴ SRA

$$\frac{V}{R} = \frac{120}{12} = 10A \qquad \therefore \quad D'I_L - \frac{V}{R} = 0$$

$$D'I_L = 0.4\cdot 25A = 10A$$

$$\frac{d}{dt}\hat{v} = \frac{-1}{RC}\hat{v} + \frac{D'}{C}\hat{i}_L - \frac{I_L}{C}\hat{d} \quad (4)$$

$$\hat{x} = \begin{bmatrix}\hat{i}_L\\ \hat{v}\end{bmatrix} \qquad \hat{u} = \begin{bmatrix}\hat{v}_s\\ \hat{d}\end{bmatrix}$$

↑ states ↑ inputs

$$\frac{d}{dt}\hat{x} = A\hat{x} + B\hat{u}$$

$$\frac{d}{dt}\begin{bmatrix}\hat{i}_L\\ \hat{v}\end{bmatrix} = \underbrace{\begin{bmatrix}0 & -D'/L\\ D'/C & -1/RC\end{bmatrix}}_{A}\begin{bmatrix}\hat{i}_L\\ \hat{v}\end{bmatrix} + \underbrace{\begin{bmatrix}1/L & V/L\\ 0 & -I_L/C\end{bmatrix}}_{B}\begin{bmatrix}\hat{v}_s\\ \hat{d}\end{bmatrix}$$

c) Find line-to-output transfer function: $G_{vg}(s)$

Standard form: $G_{vg}(s) = G_{g0} \cdot \dfrac{1}{1 + \dfrac{s}{Q\omega_0} + \dfrac{s^2}{\omega_0^2}}$

$$\begin{bmatrix} \hat{i}_L(s) \\ \hat{v}(s) \end{bmatrix} = \begin{bmatrix} G_{ig}(s) & G_{id}(s) \\ G_{vg}(s) & G_{vd}(s) \end{bmatrix} \begin{bmatrix} \hat{v}_g(s) \\ \hat{d}(s) \end{bmatrix}$$

State — Transfer functions — Input

Use $(sI-A)^{-1} \cdot B$ to get transfer functions

$$sI - A = \begin{bmatrix} s & D'/L \\ -D'/C & s + \frac{1}{RC} \end{bmatrix} = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & -D'/L \\ D'/C & -\frac{1}{RC} \end{bmatrix}$$

(a = s, b = D'/L, c = $-D'/C$, d = $s + \frac{1}{RC}$)

$$(sI-A)^{-1} = \frac{1}{ad-bc} \cdot \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \frac{1}{s^2 + \frac{1}{RC}s + \frac{D'^2}{LC}} \cdot \begin{bmatrix} s + \frac{1}{RC} & -D'/L \\ D'/C & s \end{bmatrix}$$

$$(sI-A)^{-1}B = \frac{1}{s^2 + \frac{1}{RC}s + \frac{D'^2}{LC}} \cdot \begin{bmatrix} s + \frac{1}{RC} & -D'/L \\ D'/C & s \end{bmatrix} \cdot \begin{bmatrix} 1/L & V/L \\ 0 & -I_L/C \end{bmatrix}$$

$$= \frac{1}{s^2 + \frac{1}{RC}s + \frac{D'^2}{LC}} \cdot \begin{bmatrix} \frac{s}{L} + \frac{1}{RLC} & \frac{V}{L}s + \frac{V}{RLC} + \frac{D'I_L}{LC} \\ \frac{D'}{LC} & \frac{D'V}{LC} - \frac{I_L}{C}s \end{bmatrix}$$

$$\begin{bmatrix} \hat{i}_L \\ \hat{v} \end{bmatrix} = \frac{1}{s^2 + \frac{1}{RC}s + \frac{D'^2}{LC}} \cdot \begin{bmatrix} \frac{s}{L} + \frac{1}{RLC} & \frac{V}{L}s + \frac{V}{RLC} + \frac{D'I_L}{LC} \\ \frac{D'}{LC} & \frac{D'V}{LC} - \frac{I_L}{C}s \end{bmatrix} \cdot \begin{bmatrix} \hat{v}_g \\ \hat{d} \end{bmatrix}$$

$$\frac{\hat{v}(s)}{\hat{v}_g(s)} = \frac{1}{s^2 + \frac{1}{RC}s + \frac{D'^2}{LC}} \cdot \frac{D'}{LC}$$

In standard form:

$$G_{vg}(s) = \frac{D'}{LCs^2 + \frac{L}{R}s + D'^2}$$

$$G_{vg}(s) = \frac{D'}{D'^2} \cdot \frac{1}{\frac{LC}{D'^2}s^2 + \frac{L}{D'^2R}s + 1}$$

$$G_{vg}(s) = \frac{1}{D'} \cdot \frac{1}{\frac{LC}{D'^2}s^2 + \frac{L}{D'^2R}s + 1}$$

$$\frac{1}{\omega_0^2} = \frac{LC}{D'^2} \qquad \boxed{\omega_0 = \frac{D'}{\sqrt{LC}}} \qquad f_0 = 1.111\text{kHz}$$

$$\boxed{G_{g0} = \frac{1}{D'}} = 2.5$$

$$20\log(2.5) = 7.96\text{dB}$$

$$\frac{1}{Q\omega_0} = \frac{L}{D'^2R}$$

$$\frac{\sqrt{LC}}{Q \cdot D'} = \frac{L}{D'^2R}$$

$$\frac{D'R\sqrt{LC}}{L} = \boxed{Q = \frac{D'R\sqrt{C}}{\sqrt{L}}} = 2.76$$

$$20\log(2.76) = 8.8\text{dB}$$

d) find $G_{vd}(s)$

Standard form: $G_{vd}(s) = G_{d0} \dfrac{1 - s/\omega_z}{1 + \frac{1}{Q\omega_0}s + \frac{1}{\omega_0^2}s^2}$

$$\frac{\hat{v}}{\hat{d}} = \frac{1}{s^2 + \frac{1}{RC}s + \frac{D'^2}{LC}} \cdot \left(\frac{D'V}{LC} - \frac{I_L}{C}s\right)$$

$$= \frac{\frac{D'V}{LC} - \frac{I_L}{C}s}{s^2 + \frac{1}{RC}s + \frac{D'^2}{LC}}$$

$$= \frac{1}{D'^2/LC} \cdot \frac{\frac{D'V}{LC} - \frac{I_L}{C}s}{\frac{LC}{D'^2}s^2 + \frac{L\cancel{C}}{D'^2\cancel{R}C}s + 1}$$

$$= \frac{\frac{D'V}{LC}}{\frac{D'^2}{LC}} \cdot \frac{1 - \frac{L \cdot I_L}{D'V}s}{\frac{LC}{D'^2}s^2 + \frac{L}{D'^2R}s + 1}$$

$$G_{d0} = \frac{V}{D'} = 300 = 49.5\text{dB}$$

$$\omega_z = \frac{D'V}{L \cdot I_L} \qquad f_z = 3.06\,\text{kHz}$$

$$\omega_0 = \frac{D'}{\sqrt{LC}}$$

$$\frac{1}{Q\omega_0} = \frac{L}{D'^2 R}$$

$$\frac{\sqrt{LC}}{QD'} = \frac{L}{D'^2 R}$$

$$\frac{D'R\sqrt{LC}}{L} = Q$$

$$Q = \frac{D'R\sqrt{C}}{\sqrt{L}}$$

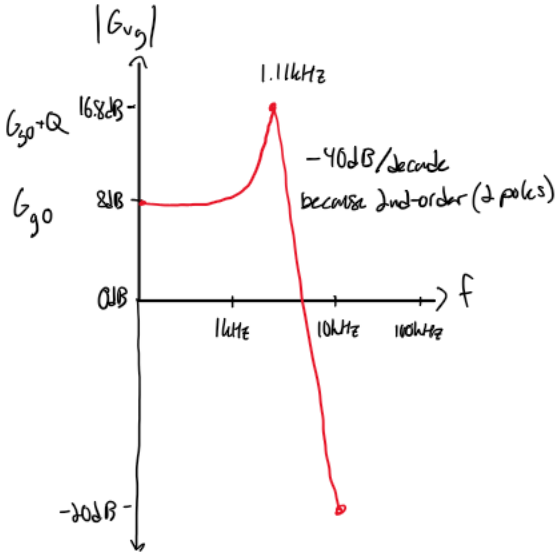
$\omega_z > 0$ ∴ Right-half plane zero

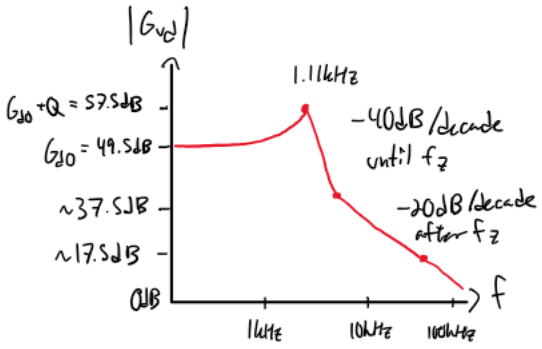
∴ Risk of instability

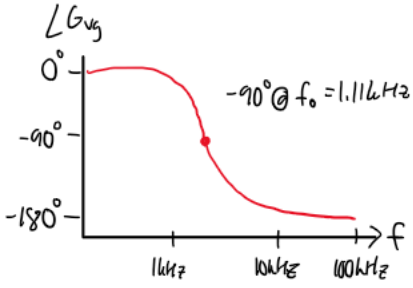
ω_0 & Q are the same for all transfer functions.

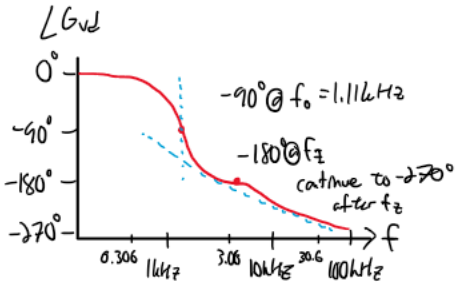
e) Sketch the magnitude & phase asymptotes for

$G_{vg}(s)$ & $G_{vd}(s)$

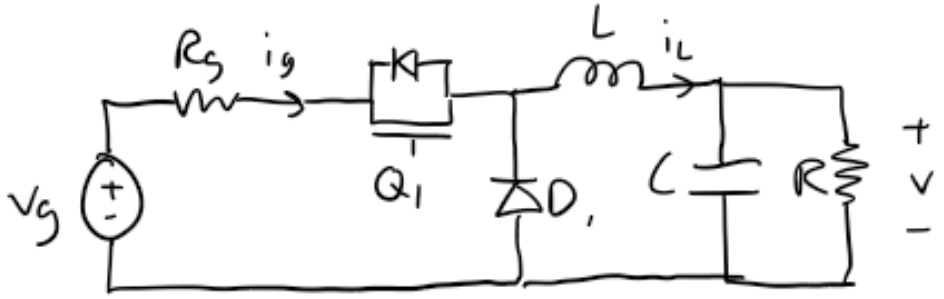








Ericksen 7.8/8.17 : Buck w/ source resistance



$V_g = 120V \quad D = 0.6 \quad R = 10\Omega$

$R_s = 2\Omega \quad L = 550\mu H \quad C = 100\mu F$

a) Average model

Given: $C\frac{d\langle v\rangle}{dt} = \langle i_L\rangle - \frac{\langle v\rangle}{R}$ ① same as normal buck

State 1 : Q_1 ON $\quad 0 < t < DT_s$

$i_g = i_L$

$V_L = \langle V_g\rangle - R_s\langle i_L\rangle - \langle v\rangle$

State 2 : Q_1 OFF $\quad DT_s < t < T_s$

$i_g = 0$

$V_L = -\langle v\rangle$

$$\langle v_L \rangle = \frac{1}{T_s} \int_0^{T_s} v_L \, dt$$

$$\langle v_L \rangle = \frac{1}{T_s} \cdot \left[\int_0^{dT_s} \langle v_g \rangle - R_g \langle i_L \rangle - \langle v \rangle \, dt + \int_{dT_s}^{T_s} -\langle v \rangle \, dt \right]$$

$$\langle v_L \rangle = \frac{1}{T_s} \cdot \left[dT_s \langle v_g \rangle - dT_s R_g \langle i_L \rangle - \cancel{dT_s \langle v \rangle} - T_s \langle v \rangle + \cancel{dT_s \langle v \rangle} \right]$$

$$\langle v_L \rangle = d\langle v_g \rangle - dR_g \langle i_L \rangle - \langle v \rangle$$

$$L \frac{d}{dt} \langle i_L \rangle = d\langle v_g \rangle - \underbrace{R_g \cdot d \cdot \langle i_L \rangle}_{\text{New}} - \langle v \rangle \quad (2)$$

Operating point: $\langle v_L \rangle = 0$

$$D \cdot V_g = D \cdot R_g \cdot I_L + V$$

$$I_L = \frac{V}{R}$$

$$D V_g = \frac{D \cdot R_g \cdot V}{R} + V$$

$$R D V_g = D R_g V + V R$$

$$R D V_g = V (R + D R_g)$$

$$V = \frac{R D V_g}{R + D R_g} = \frac{10 \cdot 0.6 \cdot 120}{10 + 0.6 \cdot 2} = \frac{720}{11.2} = 64.3 V$$

$$I_L = \frac{V}{R} = 6.43 A$$

b) Perturb & Linearize to get G_{vd} in standard form

$$L\frac{d\langle i_L\rangle}{dt} = d\langle v_S\rangle - R_S\cdot d\cdot\langle i_L\rangle - \langle v\rangle \quad (2)$$

$$\langle i_L\rangle = I_L + \hat{i}_L\,,\quad \langle v_S\rangle = V_S + \hat{v}_S\,.$$

$$d = D + \hat{d}\,,\quad \langle v\rangle = V + \hat{v}$$

$$L\cdot\frac{d\hat{i}_L}{dt} = (D+\hat{d})(V_S+\hat{v}_S) - R_S(D+\hat{d})(I_L+\hat{i}_L) - V - \hat{v}$$

$$L\cdot\frac{d\hat{i}_L}{dt} = DV_S + D\hat{v}_g + V_S\hat{d} + \hat{d}\hat{v}_S - R_S DI_L - R_S D\hat{i}_L - R_S I_L\hat{d} - R_S\hat{d}\hat{i}_L - V - \hat{v}$$

All constants should cancel out

$$DV_S - R_S DI_L - V = 0$$

$$L\cdot\frac{d\hat{i}_L}{dt} = D\hat{v}_S + V_S\hat{d} - R_S D\hat{i}_L - R_S I_L\hat{d} - \hat{v} \quad (3)$$

$$C\cdot\frac{d\langle v\rangle}{dt} = \langle i_L\rangle - \frac{\langle v\rangle}{R} \quad (1)$$

$$C\frac{d\hat{v}}{dt} = I_L + \hat{i}_L - \frac{V}{R} - \frac{\hat{v}}{R}$$

$$I_L - \frac{V}{R} = 0$$

$$C\frac{d\hat{v}}{dt} = \hat{i}_L - \frac{\hat{v}}{R} \quad (4)$$

$$\frac{d}{dt}\begin{bmatrix}\hat{i}_L\\ \hat{v}\end{bmatrix}=\begin{bmatrix}-\frac{R_S D}{L} & -1/L\\ 1/C & -1/RC\end{bmatrix}\begin{bmatrix}\hat{i}_L\\ \hat{v}\end{bmatrix}+\begin{bmatrix}D/L & \frac{V_S-R_S I_L}{L}\\ 0 & 0\end{bmatrix}\begin{bmatrix}\hat{v}_S\\ \hat{d}\end{bmatrix}$$

$$sI-A=\begin{bmatrix}s+\frac{R_S D}{L} & 1/L\\ -1/C & s+\frac{1}{RC}\end{bmatrix}$$

$$(sI-A)^{-1}=\frac{1}{\left(s+\frac{R_S D}{L}\right)\left(s+\frac{1}{RC}\right)+\frac{1}{LC}}\cdot\begin{bmatrix}s+\frac{1}{RC} & -1/L\\ 1/C & s+\frac{R_S D}{L}\end{bmatrix}$$

$$(sI-A)^{-1}B=\frac{1}{\left(s+\frac{R_S D}{L}\right)\left(s+\frac{1}{RC}\right)+\frac{1}{LC}}\cdot\begin{bmatrix}s+\frac{1}{RC} & -1/L\\ 1/C & s+\frac{R_S D}{L}\end{bmatrix}\cdot\begin{bmatrix}D/L & \frac{V_S-R_S I_L}{L}\\ 0 & 0\end{bmatrix}$$

$$=\frac{1}{s^2+\left(\frac{R_S D}{L}+\frac{1}{RC}\right)s+\frac{R_S D}{RLC}+\frac{1}{LC}}\cdot\begin{bmatrix}\frac{D}{L}\left(s+\frac{1}{RC}\right) & \left(s+\frac{1}{RC}\right)\left(\frac{V_S-R_S I_L}{L}\right)\\ D/LC & \frac{V_S-R_S I_L}{LC}\end{bmatrix}$$

$$G_{vd}(s)=\frac{\hat{v}}{\hat{d}}=\frac{(V_S-R_S I_L)/LC}{s^2+\left(\frac{R_S D}{L}+\frac{1}{RC}\right)s+\frac{R_S D}{RLC}+\frac{1}{LC}}$$

$$G_{vd}(s) = \frac{V_S - R_S I_L}{LCs^2 + \left(R_S DC + \frac{L}{R}\right)s + \frac{R_S D}{R} + 1}$$

$$G_{vd}(s) = \frac{V_S - R_S I_L}{\frac{R_S D + R}{R}} \cdot \frac{1}{\frac{LC}{\frac{R_S D + R}{R}}s^2 + \frac{\left(R_S DC + \frac{L}{R}\right)}{\frac{R_S D + R}{R}}s + 1}$$

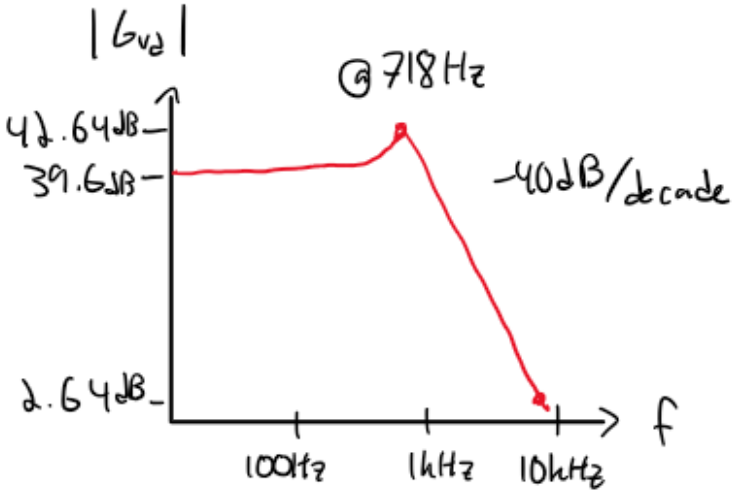
$$G_{d0} = \frac{R\left(V_S - R_S I_L\right)}{R_S D + R} = \frac{10\left(120 - 2 \cdot 6.43\right)}{2 \cdot 0.6 + 10} = \frac{1071.4}{11.2} = 95.66\,V = 39.6\,dB$$

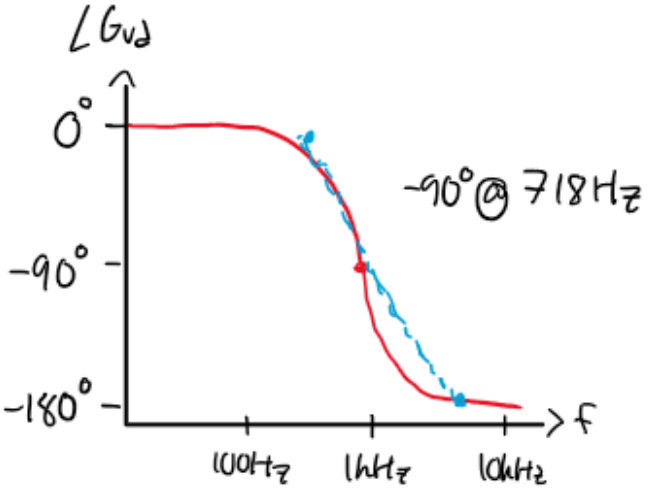
$$w_0 = \sqrt{\frac{R_S D + R}{RLC}} = \sqrt{\frac{11.2}{10 \cdot 10^{-12} \cdot 550 \cdot 10^{-6}}} = \sqrt{\frac{11.2}{550 \cdot 10^{-9}}} = 4512.6\ rad/s$$

$f_0 = 718.2\,Hz$

$$Q = \frac{R_S D + R}{R\left(R_S DC + \frac{L}{R}\right) \cdot w_0} = \frac{11.2}{10 \cdot \left(1.2 \cdot 100 \cdot 10^{-6} + \frac{550 \cdot 10^{-6}}{10}\right) \cdot 4512}$$

$$Q = \frac{11.2}{7.896} = 1.42 = 3.04\,dB$$





$$f_0 \cdot 10^{1/2Q} = 1.615kHz$$

$$f_0 \cdot 10^{-1/2Q} = 319Hz$$

If $R_S = 0$?

Q increases ∴ R_S dampens the resonant peak

$G_{dO} < \frac{V}{D}$ because there is a voltage drop across R_S, so intuitively, you need a higher duty cycle to achieve the same output voltage.